Curriculum Vitae

Piyadi Gamage Jayani Lakshika
Department of Econometrics and Business Statistics
Monash University, Australia

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GitHub: https://github.com/JayaniLakshika
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🎓 Education

PhD in Mathematics and Statistics — July 2022 — January 2026
Faculty of Business & Economics, Monash University, Australia
Thesis title: New Interactive Visual Tools and Statistical Methodology for Selecting and Evaluating Non-linear Dimension Reduction Layouts of High-dimensional Data

B.Sc. (Honours) in Statistics — January 2017 — December 2020
University of Sri Jayewardenepura (USJ), Sri Lanka
Thesis title: Computer Vision System For Automated Medicinal Plant Identification

Diploma in Computer System Design (DCSD) — March 2016 — March 2017
National Institute of Business Management (NIBM), Sri Lanka

Certificate in Business Accounting — December 2015 — September 2016
Chartered Institute of Management Accountants (CIMA), Sri Lanka

💼 Appointments Held

Research Assistant — March 2026 —
Department of Econometrics and Business Statistics, Monash University, Australia

Teaching Associate — January 2023 —
Department of Econometrics and Business Statistics, Monash University, Australia

Teaching Assistant (Temporary Instructor) — July 2021 — July 2022
University of Sri Jayewardenepura (USJ), Sri Lanka

Associate Data Analyst — January — November 2021
Vizuamatix (Pvt) Ltd, Sri Lanka

Trainee Data Analyst — July 2020 —January 2021
Vizuamatix (Pvt) Ltd, Sri Lanka

Research Internship
Vizuamatix (Pvt) Ltd, Sri Lanka
Industrial training as a fulfillment of the degree B.Sc. (Honours) in Statistics
January — July 2020

Publications

Papers

Gamage, J. P., Cook, D., Harrison, P., Lydeamore, M., & Talagala, T. S. (2025). cardinalR: Generating Interesting High-Dimensional Data Structures. arXiv preprint arXiv:2512.18172. (**Revise & Resubmit, The R Journal**)

Gamage, J. P., Cook, D., Harrison, P., Lydeamore, M., & Talagala, T. S. (2025). quollr: An R Package for Visualizing 2-D Models from Nonlinear Dimension Reductions in High-Dimensional Space. arXiv preprint arXiv:2512.18166. (**Submit, The R Journal**)

Gamage, J. P., Cook, D., Harrison, P., & Lydeamore, M. (2025). Stop Lying to Me: New Visual Tools to Choose the Most Honest Nonlinear Dimension Reduction. arXiv preprint arXiv:2506.22051 (**Revise & Resubmit, JCGS**)

Lakshika, J. P., & Talagala, T. S. (2021). Computer-aided Interpretable Features for Leaf Image Classification. arXiv preprint arXiv:2106.08077

Conference proceedings

Young Scientists' Conference on Multidisciplinary Research (YCMR) abstract publication (ISSN 2815-0260) — October 2021

17^{th} applied science research poster and abstract publication (Page 74) — September 2021

Conference presentations and contributed talks

Visualise your fitted non-linear reduction model in high-dimensional space — 2025
Australian Statistical Conference (ASC 2025) (Perth, Western Australia)

Visualise your fitted non-linear reduction model in high-dimensional space — 2025
Biometrics in the Bush Capital (BIBC2025) (Canberra, Australia)

Visualise your fitted non-linear reduction model in high-dimensional space — 2025
Monash EBS PhD Contest in International Day of Women in Statistics and Data Science (IDWSDS) (online)

Visualise your fitted non-linear reduction model in high-dimensional space — 2025
2025 Joint Statistical Meetings, Nashville, Tennessee, USA

Generating interesting high-dimensional data structures — 2025
UseR! 2025, Penn Pavilion, Duke University, Durham, North Carolina, USA

Visualise your fitted non-linear reduction model in high-dimensional space — 2024
Data Science Club, University of Nebraska, Omaha (UNO), USA

Visualise your fitted non-linear reduction model in high-dimensional space — 2024
Graphics Group, University of Nebraska, Licoln (UNL), USA

Visualise your fitted non-linear reduction model in high-dimensional space — 2024
UseR! 2024, Salzburg, Austria

Visualising how non-linear dimension reduction warps your data — 2024
Faculty of BusEco Three Minute Thesis (3MT) competition, Monash University, Australia

Visualise your fitted non-linear dimension reduction model in the high-dimensional data space — 2024
Victorian branch of the Australian and New Zealand Industrial and Applied Mathematics Society (VicANZIAM) 2024 (RMIT university, Melbourne, Australia)

Visualize your fitted non-linear dimension reduction model in the high-dimensional data space — Invited speaker — 2024
Bioinformatics Seminar, Monash University, Australia

With non-linear dimension reduction, when can you believe what you see? — 2023
Australian Statistical Conference (ASC 2023) (Wollongon, Australia)

Viewing the model from a non-linear dimension reduction in the high-dimensional data space — Invited speaker — 2023
12th-Conference of the Asian Regional Section of the International Association for Statistical Computing (IASC-ARS 2023) (Wollongon, Australia)

Higher studies webinar — Invited speaker — 2023
University of Sri Jayewardenepura (USJ), Sri Lanka

Computer Vision System for Automated Medicinal Plant Identification — Invited speaker — 2022
NUMBATs meeting, Monash University, Australia

Computer Vision System for Automated Medicinal Plant Identification — Invited speaker — 2021
Estadistica 2021, the Annual Statistics Day organized by the Statistics Society of University of Sri Jayewardenepura, Sri Lanka

Computer Vision System for Automated Medicinal Plant Identification — Invited speaker — 2021
Vx Tech Talk 2021, Vizuamatix Pvt. Ltd, Sri lanka

Statistical machine learning for medicinal plant leaves classification — Presentation — 2021
17th Applied Statistics Conference, Solvenia

Statistical machine learning for medicinal plant leaves classification — Speed-talk — 2021
Young Scientists' Conference on Multidisciplinary Research (YSCMR 2021) which is organised by the Young Scientists' Association of the National Institute of Fundamental Studies, Sri Lanka (NIFS-YSA)

Visiting

Prof. Heike Hofmann, A/Prof. Susan Vanderplas, and Dr. Michelle Graham — October-December 2024
At the University of Nebraska, Lincoln, USA (six weeks)

A/Prof. Ursula Laa — July 2024
At the University of Natural Resources and Life Sciences, Vienna (BOKU University), accompanied by Prof. Di Cook, Prof. Eun-Kyung Lee, and A/Prof. Natalia da Silva (one week)

R packages

airpurifyr: Provide an interface to scrape air quality data from https://openaq.org/

cardinalR: Collection of data structures.

quollr: Tool to visualise how non-linear dimension reduction warps your data

polarisR: Shiny app to compare models generated with `quollr` in high dimensions using different tour methods, featuring interactivity.

olympics: Provide an interface to scrape Olympic data from https://olympics.com/

MedLEA: Software which contains information about medicinal plants in Sri Lanka

DSjobtracker: Software which contains what skills and qualifications are required for data science related jobs

Teaching

Tutor

[ETC2440] **Mathematics for economics and business**	2026
[ETC5242] **Statistical thinking**	2025
[ETC2420] **Statistical thinking**	2025
[ETC5510] **Introduction to data analysis**	2024, 2026
[ETC1010] **Introduction to data analysis**	2024
[ETC3250] **Introduction to machine learning**	2023 — 2024, 2026
[ETC5250] **Introduction to machine learning**	2023 — 2026
[ETC5521] **Exploratory data analysis**	2023
[STA 326 2.0] **R Programming**	2022
[STA 213 2.0] **Inferential Statistics**	2022
[STA 214 1.0] **Nonparametric Statistics**	2022
[STA 215 2.0] **Sampling techniques**	2022
[STA 115 1.0] **Elements of Sampling**	2022
[STA 113 2.0] **Descriptive Statistics**	2022
[STA 114 2.0] **Probability and Distribution Theory I**	2022
[STA 224 2.0] **Regression Analysis**	2021
[STA 225 2.0] **Design of Experiments**	2021
[STA 226 1.0] **Data Analysis II**	2021
[STA 123 2.0] **Probability and Distribution Theory II**	2021
[STA 124 2.0] **Data Analysis I**	2021

[STA 125 1.0] **Statistical Communication** 2021

Workshops

Reproducible Reporting and Research with Quarto September 2025

Reproducible Reporting, Academic Papers, Presentations and Theses with Quarto July 2025

Awards

PhD external experience fund 2024
Monash University, Australia

Monash Business School Prestigious International Conference Award 2024
Monash University, Australia

Co-Funded Graduate Research Scholarship, Monash Business School 2022 — 2026
Monash University, Australia

Diploma in Computer System Design (DCSD) — Gold medalist 2017
National Institute of Business Management (NIBM), Sri Lanka

OCTAVE competition — Finalists 2021
Research abstract competition organized by OCTAVE company, Sri Lanka

Stat Circle quiz competition — Winners 2018
Team member of the winners at inter university quiz competition organized by the Stat Circle Faculty of Science, University of Colombo, Sri Lanka

Academic Service & Community Engagment

Monash Innovation Guarantee-Undergraduate (MIG-UG)— Mentor January 2026
Monash University, Australia

ASC Session Chair 2025
Australian Statistical Conference in Perth, Australia

Monash Innovation Guarantee-Postgraduate (MIG-P)— Mentor June 2025
Monash University, Australia

NUMBATs seminar organizer 2025
Monash University, Australia

UseR! 2024 Session Chair 2024
Conference in Salzburg, Austria

ASC Session Chair 2023
Australian Statistical Conference in Wollongong, Australia

WOMBAT tutorials — Helper 2022
Monash University, Australia

Statistical Consultancy Service — Department representative 2020
University of Sri Jayewardenepura (USJ), Sri Lanka

Professional Memberships

American Statistical Association (ASA) 2025–

Statistical Society of Australia (SSA) 2022–

R-Ladies Melbourne 2022–

Statistics society 2017–2022
University of Sri Jayewardenepura (USJ), Sri Lanka

References

Prof Dianne Cook
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